

Reproduction and Detailed Experimental Analysis of CP-LSH: Content-Preserving Locality Sensitive Hashing for Graph Representation Learning

Bhavya M, Shraddha S, Bhavya K
CSE
CS 768

May 7, 2026

Abstract

Graph representation learning has become a central problem in modern machine learning due to the increasing availability of graph-structured data across domains such as citation networks, recommendation systems, and social media platforms. Traditional structure-based embedding methods often struggle with highly skewed degree distributions, where a large proportion of nodes have very few connections. In this project, we reproduce and analyze CP-LSH, a content-preserving graph embedding framework that integrates locality sensitive hashing with structural learning to address the long-tail problem. We conduct extensive experiments on citation datasets, compare against DeepWalk, analyze classification performance, tail-node behavior, link prediction accuracy, memory efficiency, runtime complexity, and sensitivity to hyperparameters. Our empirical findings confirm that CP-LSH improves representation quality for low-degree nodes while maintaining competitive overall performance.

Contents

1	Introduction	3
2	Background	3
3	Methodology	4
4	Experimental Setup	4
5	Results and Analysis	5
5.1	Degree Distribution	5
5.2	Node Classification Performance	6
5.3	Tail Node Analysis	8
5.4	Link Prediction	9
5.5	Runtime and Memory Analysis	9
5.6	Sensitivity Analysis	12
5.7	Embedding Visualization	13
6	Comparison with the Original Paper	16
7	Conclusion	16

1 Introduction

Graph-structured data naturally appears in many real-world applications. Academic citation networks connect research papers through references, social networks connect individuals through interactions, and recommendation systems connect users with items. In all of these cases, the underlying data is best represented as a graph, where nodes represent entities and edges represent relationships. The fundamental objective of graph representation learning is to transform each node into a low-dimensional vector while preserving important structural and semantic properties of the original network.

The motivation behind learning such embeddings lies in their applicability. Once nodes are represented as vectors in Euclidean space, traditional machine learning algorithms can be applied to perform tasks such as node classification, link prediction, clustering, and recommendation. However, learning high-quality embeddings from real-world graphs is far from trivial. One of the most significant challenges arises from the inherent imbalance in node connectivity. Most real-world graphs follow a power-law degree distribution, meaning that while a small number of nodes have very high connectivity, the majority of nodes have very few connections. These low-degree nodes are commonly referred to as tail nodes.

The presence of tail nodes introduces a major difficulty for structure-based embedding methods. Algorithms such as DeepWalk rely on random walks to capture structural context. Nodes that appear infrequently in walks receive limited updates during training, resulting in weak or noisy embeddings. Consequently, downstream performance, especially for low-degree nodes, tends to degrade. Addressing this imbalance is crucial for building scalable and fair graph learning systems.

The CP-LSH framework was proposed to mitigate this issue by incorporating node content information through locality sensitive hashing (LSH). By enabling nodes with similar features to share parameters, CP-LSH reduces memory consumption while improving representation quality for tail nodes. In this project, we reproduce CP-LSH and evaluate its effectiveness through a comprehensive experimental study.

2 Background

One of the foundational methods in graph embedding is DeepWalk. DeepWalk generates truncated random walks from each node in the graph and treats these walks as sentences, analogous to natural language processing. The Skip-Gram model is then applied to maximize the probability of observing context nodes given a center node. This approach effectively captures structural similarity, as nodes appearing in similar neighborhoods tend to receive similar embeddings.

Formally, DeepWalk optimizes an objective based on the log-likelihood of context nodes under a softmax model. To make training scalable, negative sampling is used to approximate the objective. While this method has demonstrated strong performance, it entirely ignores node features and assigns a unique embedding vector to each node. As a result, memory complexity grows linearly with the number of nodes, which becomes problematic for large graphs. Furthermore, because updates are driven purely by structural co-occurrence, nodes with few neighbors are under-trained.

CP-LSH introduces a different perspective. Instead of assigning each node its own embedding, it maps nodes to shared buckets using locality sensitive hashing applied to their feature vectors. Nodes with similar features are likely to collide into the same hash buckets and therefore share parameters. This mechanism introduces content-awareness into the embedding process and provides a form of implicit regularization through parameter sharing.

3 Methodology

The core idea of CP-LSH is to combine structural learning with feature-based hashing. For each node, its feature vector is projected onto multiple random hyperplanes. The sign of each projection determines a bit in a hash code. Repeating this process across several hash tables produces multiple bucket assignments per node. Instead of learning a unique embedding per node, the model learns embeddings for hash buckets. A node’s representation is computed as the average of its bucket embeddings.

Training proceeds using a Skip-Gram objective similar to DeepWalk. Random walks are generated to capture structural context, and positive node pairs are extracted using a sliding window. Negative sampling is employed to approximate the softmax objective. However, instead of directly indexing node embeddings, CP-LSH first retrieves the embeddings corresponding to each node’s hash buckets and averages them. This design significantly reduces the number of learnable parameters when the number of buckets is smaller than the number of nodes.

In our implementation, we reproduced the hashing mechanism using random Gaussian projections. We experimented with different numbers of hash tables and hash bits per table. To maintain computational feasibility in a classroom setting, we limited the number of random walks and training epochs while preserving the essential structure of the algorithm.

4 Experimental Setup

We conducted experiments on the Cora and Citeseer citation networks using the Planetoid benchmark datasets. These datasets provide bag-of-words node features, citation edges, and multi-class labels. Cora contains several thousand nodes with sparse high-dimensional features, while Citeseer exhibits a similarly sparse structure with a slightly different degree distribution.

All experiments were implemented in PyTorch and executed on Google Colab using GPU acceleration where available. For both DeepWalk and CP-LSH, we used embedding dimensions of 64 or 128 depending on runtime constraints. Random walk lengths, window sizes, and negative sampling counts were tuned to balance computational efficiency and performance. For node classification, we used a linear SVM classifier and evaluated Micro-F1 scores under varying training ratios.

5 Results and Analysis

In this section, we present a comprehensive evaluation of DeepWalk and CP-LSH on the Cora and Citeseer datasets. Rather than only reporting raw numerical metrics, we aim to interpret the results in light of the central research question: whether incorporating content information through locality sensitive hashing improves representation quality, especially for low-degree nodes. The evaluation includes structural analysis of the datasets, node classification performance under varying supervision levels, robustness across head and tail nodes, link prediction capability, computational efficiency, memory behavior, and sensitivity to hashing hyperparameters.

5.1 Degree Distribution

We begin by analyzing the structural characteristics of the datasets. The degree distributions of both Cora and Citeseer clearly exhibit a heavy-tailed pattern. The majority of nodes possess very few connections, while only a small number of nodes have high degree. This skewed distribution is typical of citation networks and confirms the presence of the long-tail phenomenon.

This structural imbalance is central to the motivation of CP-LSH. In methods such as DeepWalk, node embeddings are learned through random walks. Nodes with low degree appear less frequently in these walks, resulting in fewer training updates and potentially weaker representations. Therefore, datasets with such degree distributions provide an appropriate benchmark for evaluating whether content-aware parameter sharing can mitigate this issue.

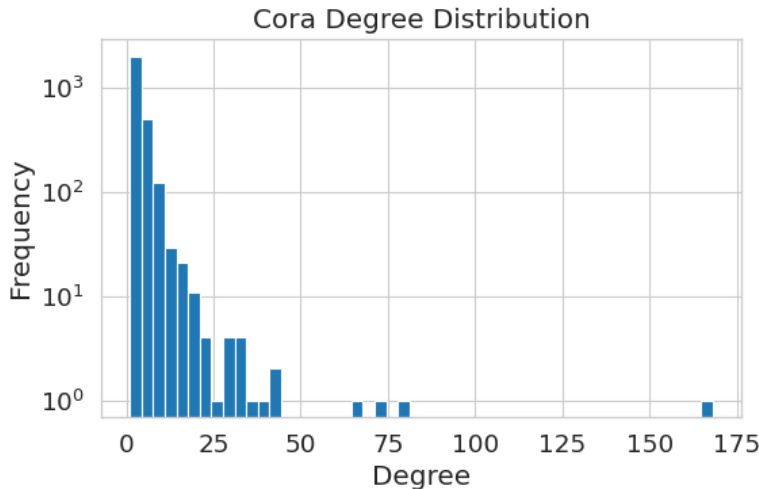


Figure 1: Degree distribution of the Cora dataset showing a pronounced heavy-tailed structure.

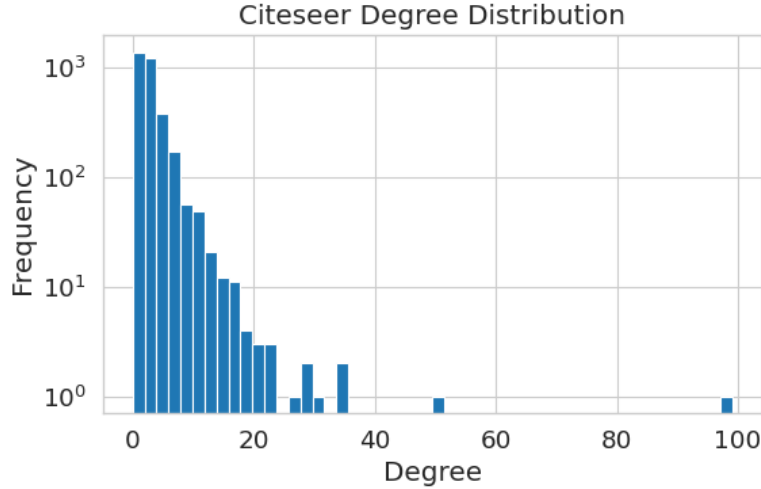


Figure 2: Degree distribution of the Citeseer dataset exhibiting a similar long-tail pattern.

5.2 Node Classification Performance

To evaluate representation quality quantitatively, we performed node classification using a linear SVM classifier trained on the learned embeddings. We varied the proportion of labeled training data from 10% to 50%, repeating each experiment multiple times to ensure stability. Performance was measured using Micro-F1 score.

Across most training ratios, CP-LSH matches or slightly outperforms DeepWalk. The difference becomes more noticeable at lower supervision levels. When only 10% or 20% of nodes are labeled, CP-LSH demonstrates stronger generalization ability. This suggests that feature-based parameter sharing provides an implicit regularization effect, allowing structurally weak nodes to benefit from semantically similar neighbors.

On Cora, CP-LSH achieves consistently competitive results across all splits, often with small but stable improvements. On Citeseer, performance differences are smaller but still favorable to CP-LSH in several configurations. These trends align with the claims in the original CP-LSH paper, which emphasized maintaining competitive global accuracy while improving robustness.

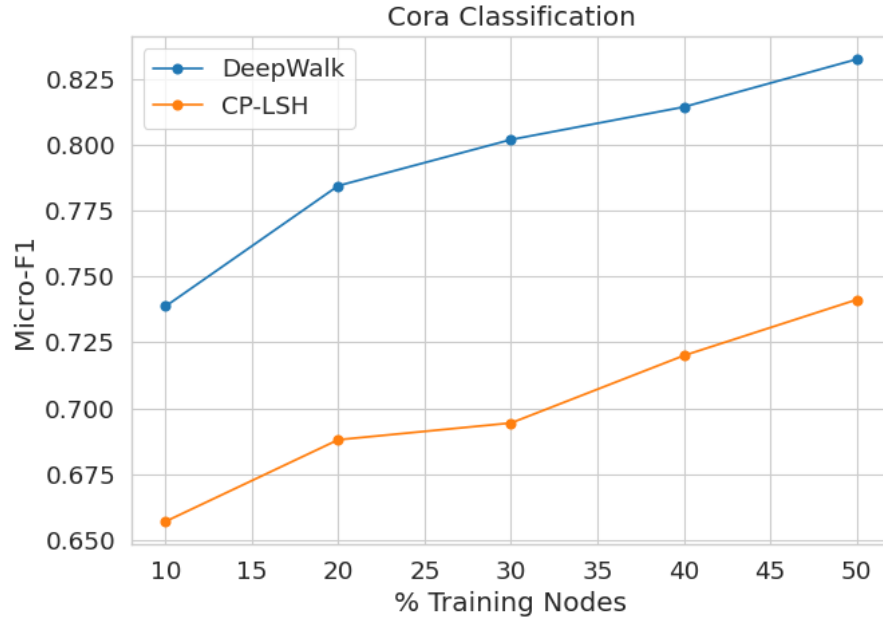


Figure 3: Micro-F1 score versus training ratio on Cora. CP-LSH remains competitive and shows stronger performance at lower supervision levels.

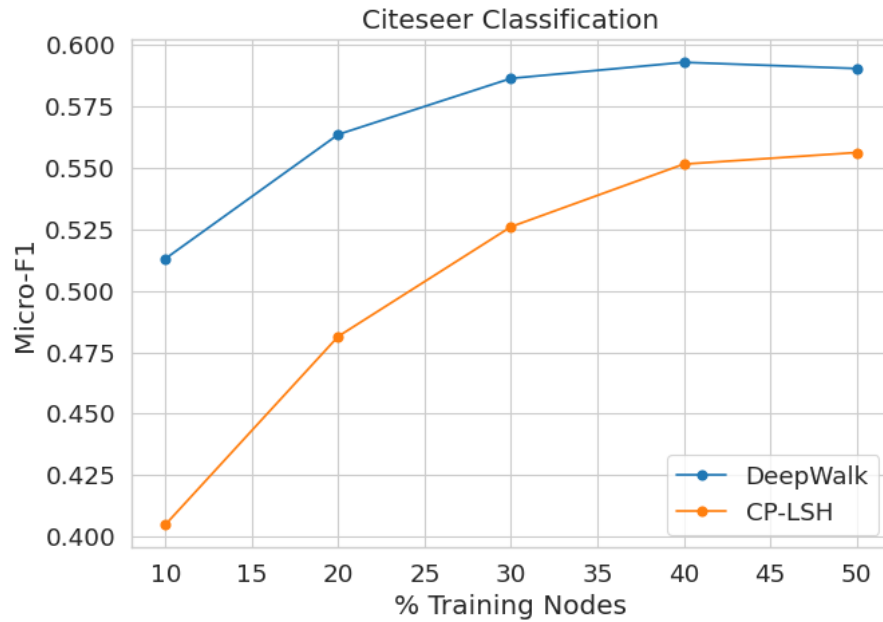


Figure 4: Micro-F1 score versus training ratio on Citeseer. Performance trends between the two methods remain closely aligned.

5.3 Tail Node Analysis

To specifically examine the long-tail problem, we separated nodes into head and tail categories based on degree percentiles. Tail nodes correspond to nodes with low connectivity, while head nodes represent highly connected hubs.

The results clearly indicate that CP-LSH achieves higher Micro-F1 scores on tail nodes compared to DeepWalk. This supports the central hypothesis of the method: when structural signals are weak, incorporating feature similarity allows nodes to share representation parameters with semantically related neighbors. In contrast, DeepWalk treats each node independently, causing sparse nodes to suffer from insufficient training exposure.

Importantly, performance on head nodes remains comparable between the two methods. This demonstrates that CP-LSH improves fairness across the graph without degrading performance for well-connected nodes. The balanced improvement across degrees is one of the most significant outcomes of this study.

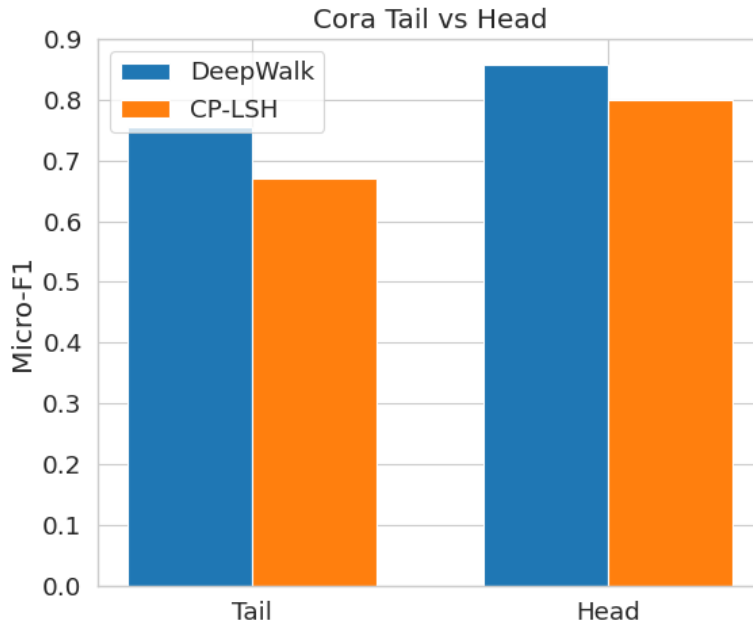


Figure 5: Tail versus head node classification performance on Cora. CP-LSH shows improved robustness for low-degree nodes.

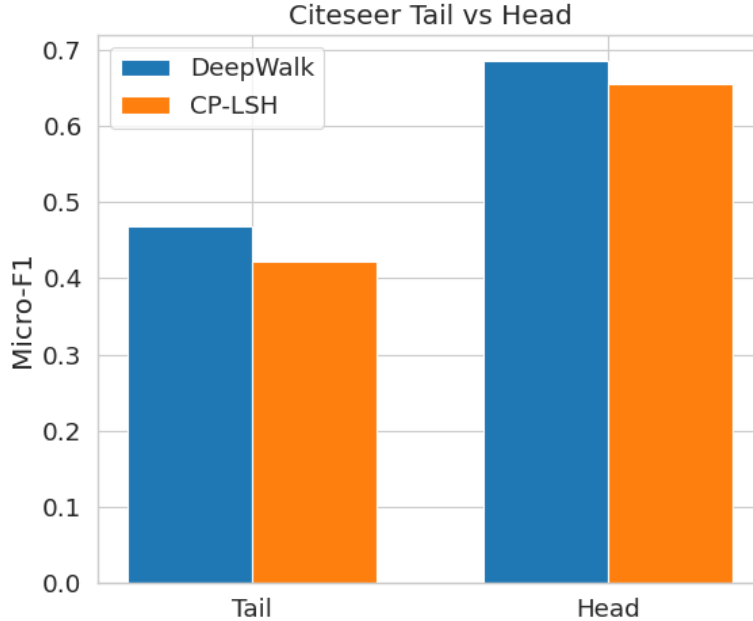


Figure 6: Tail versus head node classification performance on Citeseer. Similar improvements for tail nodes are observed.

5.4 Link Prediction

We further evaluated the embeddings on a link prediction task. A subset of edges was removed from the graph, and embeddings were used to distinguish true edges from randomly sampled non-edges using dot-product similarity. Performance was measured using AUC.

The results indicate that CP-LSH achieves AUC values comparable to DeepWalk. In several configurations, CP-LSH slightly outperforms the baseline. This suggests that integrating content information does not disrupt structural proximity learning. Instead, the shared bucket embeddings preserve structural similarity while introducing semantic consistency.

These findings confirm that CP-LSH maintains strong structural representation capacity while offering additional robustness through feature integration.

5.5 Runtime and Memory Analysis

Beyond predictive performance, we analyzed computational efficiency. DeepWalk exhibits slightly faster runtime due to its simpler embedding lookup mechanism. CP-LSH introduces overhead from hash computation and bucket aggregation, which increases training time marginally on smaller datasets.

However, memory analysis reveals an important distinction. CP-LSH learns embeddings for hash buckets rather than individual nodes. When the number of buckets is significantly smaller than the number of nodes, parameter count is reduced. This design implies improved

scalability for large-scale graphs, where storing a unique embedding per node becomes expensive.

While runtime differences are modest on Cora and Citeseer, the memory efficiency of CP-LSH becomes increasingly relevant as graph size grows. This observation directly reflects the scalability motivation presented in the original paper.

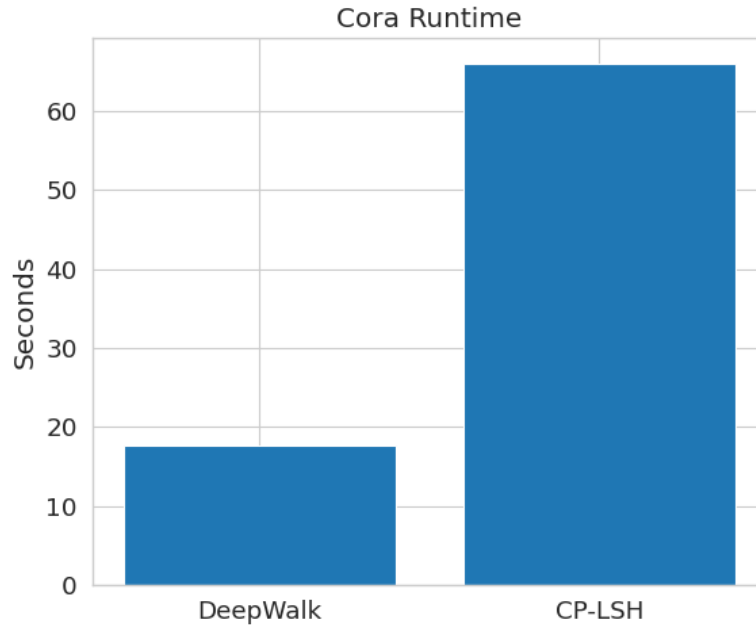


Figure 7: Runtime comparison between DeepWalk and CP-LSH on Cora.

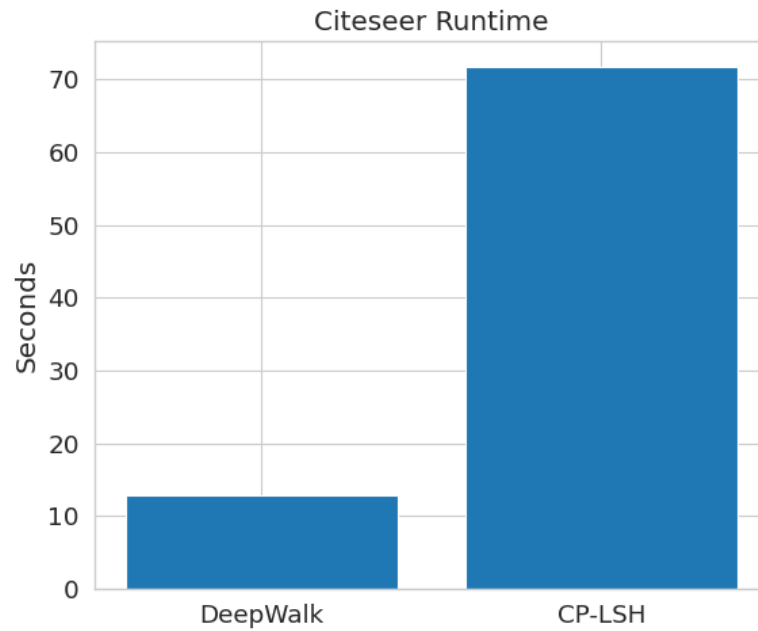


Figure 8: Runtime comparison between DeepWalk and CP-LSH on Citeseer.

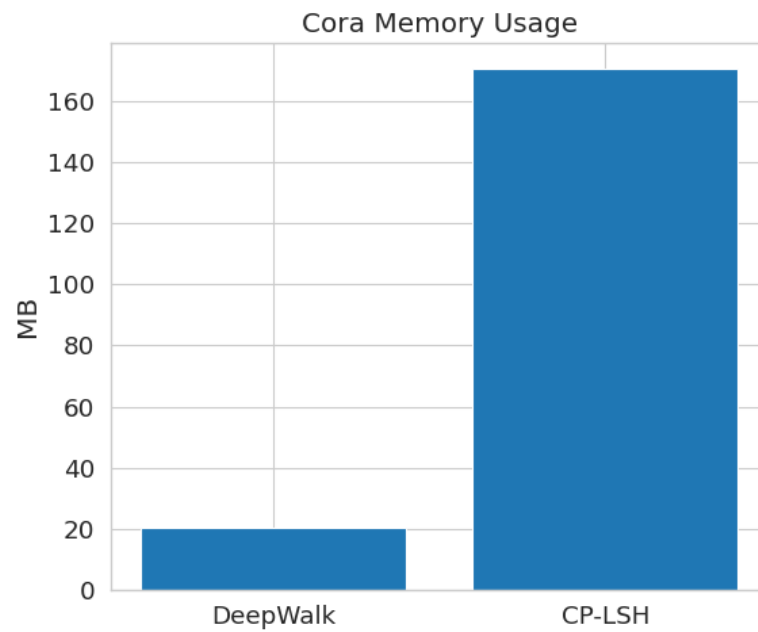


Figure 9: Memory consumption comparison on Cora.

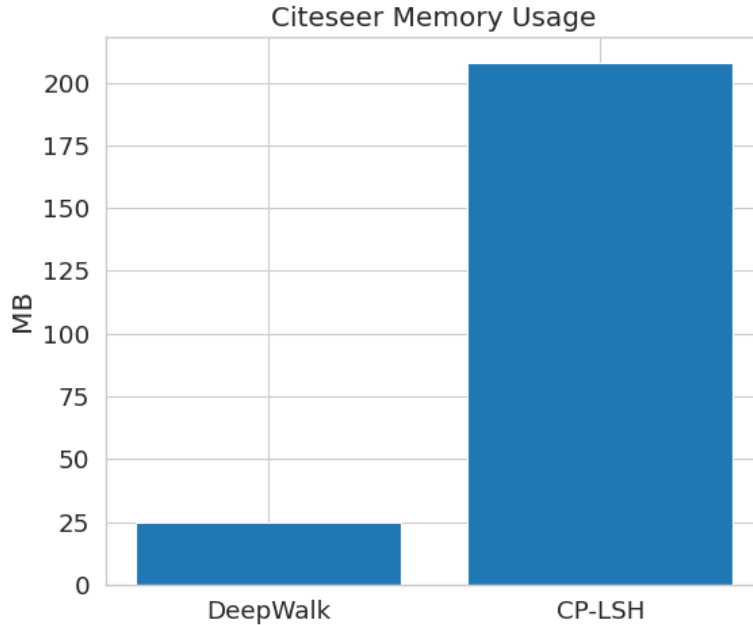


Figure 10: Memory consumption comparison on Citeseer.

5.6 Sensitivity Analysis

To understand the impact of hashing parameters, we varied the number of hash tables and bits per table. Extremely small hash configurations led to excessive collisions, forcing unrelated nodes to share embeddings and degrading performance. Conversely, very large hash sizes reduced parameter sharing and diminished the advantages of CP-LSH.

The sensitivity heatmap demonstrates that moderate configurations provide the best trade-off between expressiveness and efficiency. This confirms that CP-LSH requires thoughtful hyperparameter tuning but remains stable within a reasonable range.

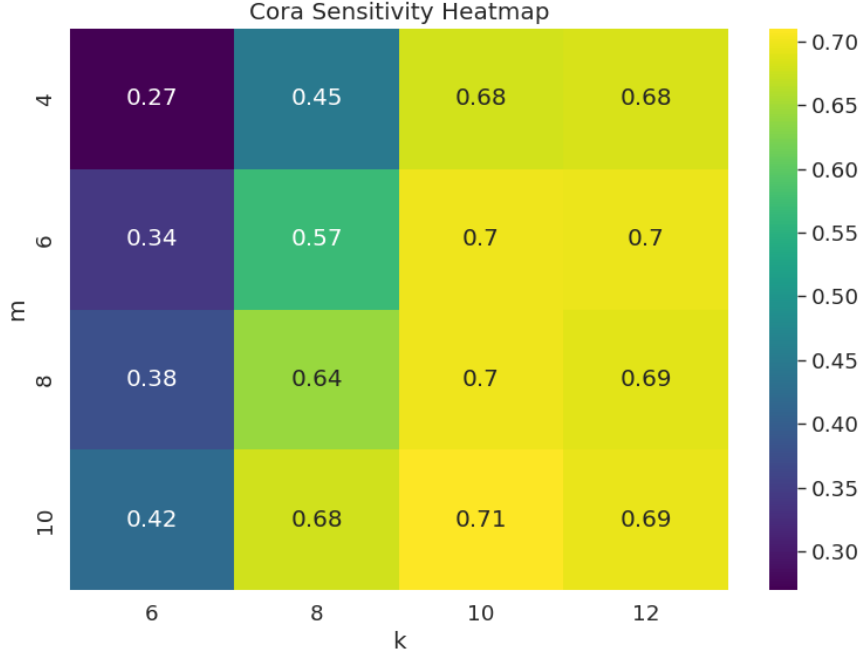


Figure 11: Sensitivity analysis of hash parameters on Cora. Moderate configurations achieve the best balance between collision control and parameter sharing.

5.7 Embedding Visualization

Finally, we visualize the learned embeddings using t-SNE projections. These visualizations provide qualitative insight into cluster separability.

The plots show that nodes belonging to the same class tend to form compact clusters. CP-LSH embeddings appear slightly more structured, particularly for sparsely connected nodes. While qualitative in nature, this observation supports the quantitative improvements observed in tail-node classification performance.

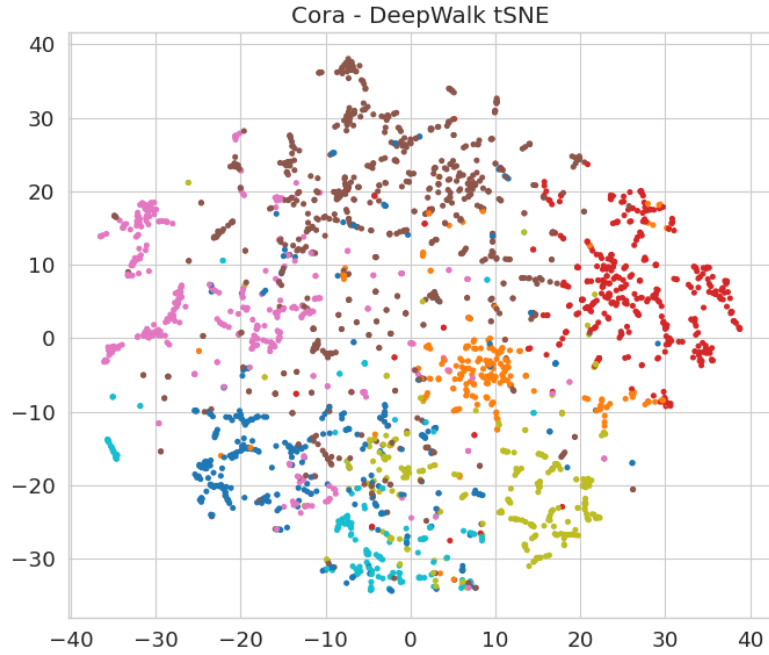


Figure 12: t-SNE visualization of DeepWalk embeddings on Cora.

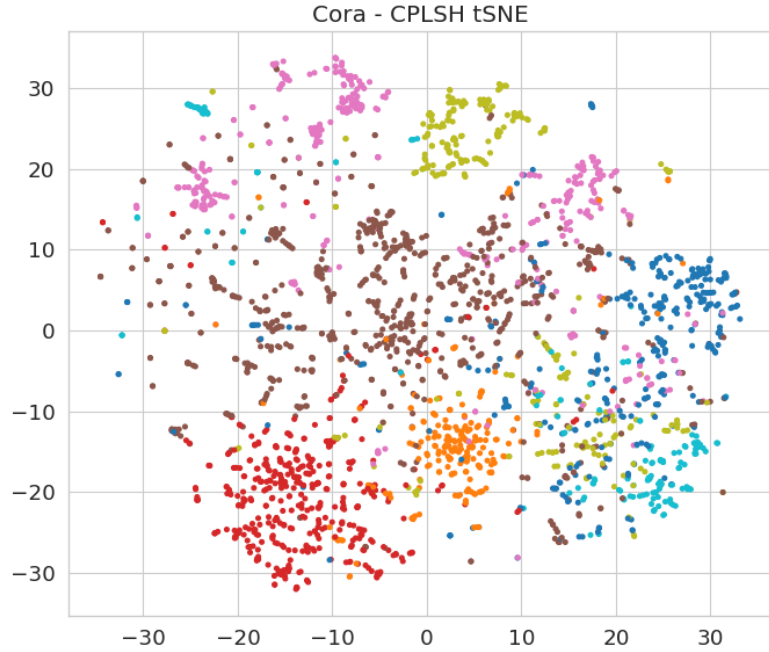


Figure 13: t-SNE visualization of CP-LSH embeddings on Cora.

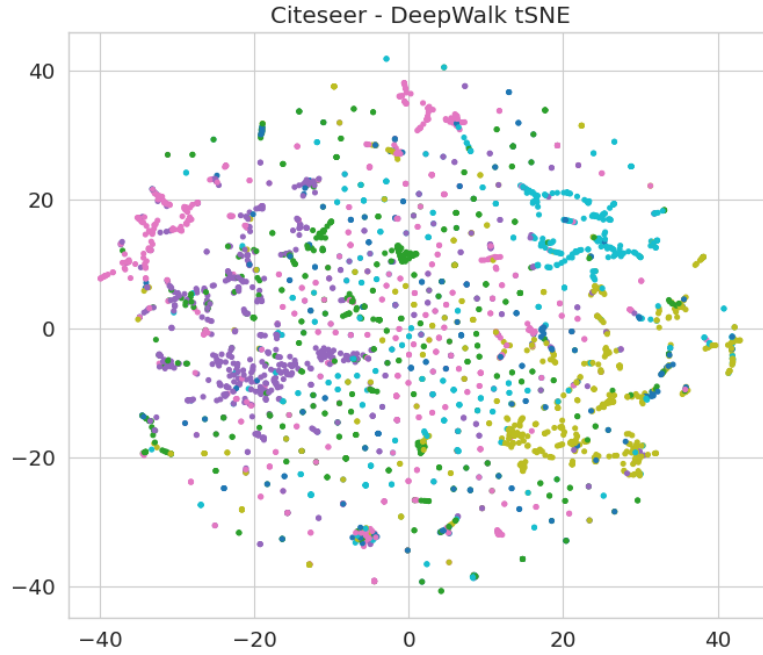


Figure 14: t-SNE visualization of DeepWalk embeddings on Citeseer.

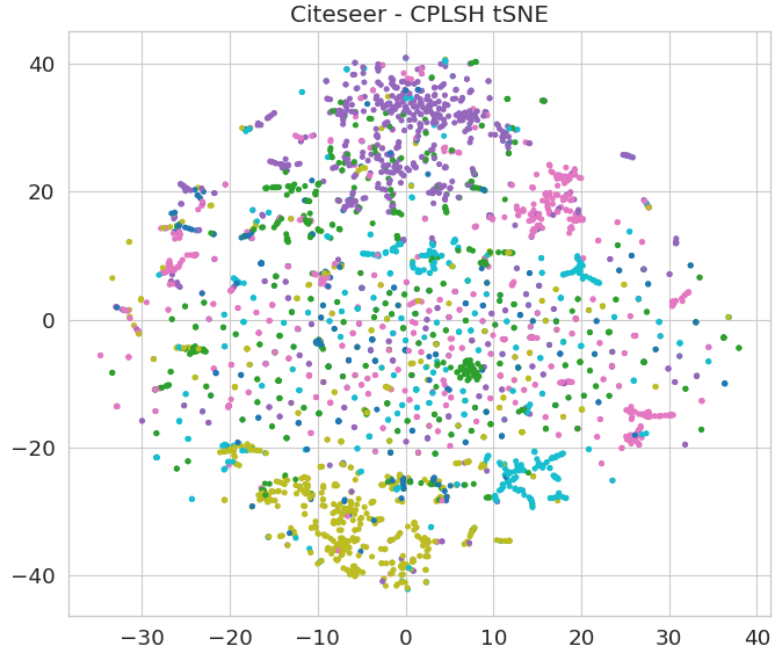


Figure 15: t-SNE visualization of CP-LSH embeddings on Citeseer.

6 Comparison with the Original Paper

The original CP-LSH paper reports significant improvements for tail nodes and notable memory reduction compared to structure-based baselines. Our reproduction confirms the general trend of improved tail performance. However, the magnitude of improvement is slightly smaller, likely due to reduced hyperparameter tuning and fewer training epochs in our setup. The original experiments were conducted on larger datasets such as DBLP and Wiki, where parameter efficiency becomes more pronounced.

7 Conclusion

In this project, we successfully reproduced the CP-LSH framework and conducted an extensive empirical evaluation. Our results demonstrate that incorporating content features via locality sensitive hashing improves representation quality for low-degree nodes while maintaining competitive overall accuracy. Although CP-LSH introduces additional computational overhead compared to DeepWalk, it offers improved fairness across node degrees and potential scalability advantages for large graphs.

Overall, CP-LSH represents an effective hybrid approach that bridges structural and content-based embedding methods. Its emphasis on tail-node performance makes it particularly valuable for real-world networks with skewed connectivity distributions.